helix_proxy — VPN-LAN Service Access Feature Status

Revision: 2 **Last modified:** 2026-07-01T19:15:00Z **Status:** Feature COMPLETE (Phases 0-12 on `main`). Anti-bluff bridge scaffold + preflight doctor + operator guide + Challenge runner + **Phase-1 SSRF carve-out teeth + Phase-12 bidirectional ingress-allowlist teeth** + the **§11.4.135 autonomous battery** (stress+chaos, benchmark+memory, concurrency+load, DNS-rebinding gap) are PROVEN NOW (autonomously, no live VPN, no podman) and wired GREEN into the standing suite. Every live protocol round-trip (SMB/NFS/FTP/SFTP/WebDAV/email/Cast/DIAL/ADB) is AUTHORED with real byte-integrity / sink-side evidence gating **plus a Phase-12 both-way reverse-leg assertion** (NFS NLM/NSM callback, FTP active, Cast callback, adb reverse, email N/A §11.4.6) and honestly SKIPs while the svord bridge is down (§11.4.3) — never a fake PASS; Miracast is Won't-fix structurally-impossible (§11.4.112). Operator-gated: the svord bridge (secrets + Mullvad + root/sudo) → unlocks all live round-trips BOTH directions; the multicast-reflector remote deployment; any real-device flash. **Data-plane env-block (operator-actionable, NOT a code defect):** host rootless-podman `aardvark-dns` cannot bind netavark `:53` → `:53128`/`:51080` don't serve + `./start` fails; the standing suite honestly reports this (3 env FAILs) rather than hanging. **Authority:** Inherits `constitution/Constitution.md` per §11.4.35. This is the §11.4.153 per-feature Status set (companion §11.4.56 two-audience `Status_Summary.md`) for the VPN-LAN service-access feature workstream (`feature/vpn-aware-dynamic-routing`, §11.4.167). Every row is reconciled against real committed code + tests + design docs; no metadata-only PASS and no fabricated video-confirmation (§11.4 / §11.4.1 / §11.4.6). **Companion / cross-refs:** integration status `../../design/vpn_lan_access/Status.md` · plan `../../design/vpn_lan_access/PLAN.md` · Miracast verdict `../../design/vpn_lan_access/miracast_verdict.md` · reflector design `../../design/vpn_lan_access/reflector_design.md` · operator guide `../../guides/vpn_lan_bridge_setup.md`. **Repo anchor:** branch `main` (all work merged to + performed on main per operator directive). Landing commits: bridge scaffold `d781002` + standing-suite wiring `5c28f56`; Phase 2/3 file protocols `182e80a`; Phase 4 email `2f31460`; Phase 5 reflector `d0b42df`; Phase 6/7 Cast+ADB `65043ce`; Phase 8 Miracast `12faf12`; Phase 9 guide `a5e5616`; Phase 11 Challenge runner + HelixQA bank `89f73b7`; Phase 1 SSRF teeth `1d87b3a`; Phase 10 containerization `1911d5e`; Phase 12 bidirectional `2ed0fed` + both-way protocol assertions `cc8b620`; stress+chaos `97d9733`; bench+memory + dns-rebinding `4e2810c`; concurrency+load `7e73b08`; §11.4.135 suite guards `6d8893f`.

Operator-blocked / operator-gated items (read first — §11.4.45 O(1) surface)

Every autonomous claim below is either PROVEN NOW against a committed file/evidence artefact, or an honest SKIP that flips to a live PASS the moment the operator supplies the input named here. No live round-trip is marked PASS without captured evidence (§11.4.6 / §11.4.69).

Item	Why blocked / gated	Unblock condition
Live protocol round-trips (SMB / NFS / FTP / FTPS / SFTP / WebDAV / IMAP / SMTP-submission / POP3S / Cast / DIAL / ADB)	Need the live svord bridge — secrets + Mullvad WireGuard + root/sudo on the bridge side (§11.4.21 / §11.4.10). The scaffold proves the honest-SKIP path + the anti-bluff gate NOW; the live byte/sha256/mailbox/serial evidence exists only when the bridge is genuinely up. Bridge-down ⇒ honest SKIP (`network_unreachable_external`), never a fake PASS (§11.4.3 / §11.4.69).	Operator copies `.env.example`→`.env`, points the 6-var contract at `svord_toolkit`, brings the VPN up; each test then emits real captured round-trip evidence under `qa-results/vpn_lan/phase<N>/<ts>/`.
Multicast discovery reflector — remote deployment	Deploying an Avahi mDNS / SSDP-relay reflector changes a remote host (§11.4.122) — requires an interactive keep/deploy decision. The reflector *design* + the client-side enumeration *test* are committed and autonomous; only the remote deploy is gated.	Operator authorizes the remote-side reflector deployment (Phase 10 boots it on-demand via the containers submodule, §11.4.76).
ADB flash over VPN (`usbip` USB-over-IP)	Flashing a real device is high-blast-radius (§11.4.133 target-hardware-safety) and USB-level, not network-routable (fastboot is USB-bound — recon 4 FACT). The test NEVER flashes; it	Operator authorizes the specific device flash via a remote host with the device attached over `usbip`.

Item	Why blocked / gated	Unblock condition
	records the honest boundary and SKIPs `operator_attended`.	
HelixQA `vpn_lan.yaml` autonomous session	The `helixqa` binary cannot be built in this checkout — its `go.mod replaces` 6 un-vendored own-org sibling modules (§11.4.3). The bank is authored; no autonomous HelixQA session has been run and none is claimed. The runnable proof NOW is the Challenge runner.	Vendor the 6 sibling modules under `submodules/`, then drive the bank via an autonomous HelixQA session.

Feature status table (§11.4.153)

Legend. *Wiring* = is the feature genuinely reachable/usable by an end user of the deployed helix_proxy through the env-var bridge (§11.4.108)? *Validation* ∈ {PASS, FAIL, SKIP, PENDING_FORENSICS, OPERATOR-BLOCKED}. A **PASS** cites a committed captured artefact or a self-proving standing-suite gate. **SKIP** = honest topology/precondition skip (bridge down / reflector not deployed / server absent — §11.4.3), never folded into a PASS. *Video-confirmation* is **N/A** for every row: helix_proxy VPN-LAN is a headless network feature with no rendered UI — its §11.4.107/§11.4.69 proof is captured exit codes + byte-integrity sha256 + sink-side protocol responses (SMB/NFS/mailbox/eureka JSON/device serial), not a rendered video (§11.4.153(3) autonomous-infeasible ⇒ honest non-video evidence). Live-path rows additionally carry no autonomous video because the live round-trip needs the operator-gated svord bridge.

A. Bridge foundation & anti-bluff scaffold — PROVEN NOW (autonomous, no live VPN)

Component	Feature	Category	Implementation
env-var bridge contract	6-var decoupled bridge contract (`HELIX_SVORD_DIR` / `_CONNECT` / `_DISCONNECT` / `_HEALTH` / `_SUBNET` / `_HOST`); `bridge_load` / `bridge_up` / `bridge_require` primitives (rc 2 = OPERATOR-BLOCKED, rc 3 = misconfigured)	foundation / decoupling (§11.4.28)	`.env.example` (tracked, no secrets); `tests/lib/svord_bridge.sh`

Component	Feature	Category	Implementation
svord-doctor preflight	Deterministic 3-verdict preflight — `BRIDGE: UP` (exit 0) / `SKIP:network_unreachable_external` (exit 2) / `MISCONFIGURED:<reason>` (exit 3): env resolvable? hooks executable? `HELIX_BRIDGE_HEALTH` exit 0? remote host reachable? Never fakes UP	preflight / anti-bluff gate	`scripts/svord_doctor.sh`; companion `docs/scripts/svord_doctor.md` (§11.4.18)

B. File shares — AUTHORED, live round-trip operator-gated

Component	Feature	Category	Implementation	Wiring (reachal
SMB / CIFS / NMB	SMB/CIFS write→read-back→sha256 round-trip over the L3-routed VPN share (`smbclient`, `mount.cifs` fallback); NMB name-resolution by 10.x unicast (multicast NMB not routed)	file-share / functional	`tests/vpn_lan/smb_nfs_roundtrip.sh` (SMB slice)	Gated — reachabl the bridg + `HELIX_VPN` set; route L3, NOT SOCKS5
NFS	NFS write→read-back→sha256 round-trip over the VPN export (pre-mounted `HELIX_VPN_NFS_MOUNTED`, or root temp-mount `HELIX_VPN_NFS_EXPORT`); temp mount	file-share / functional	`tests/vpn_lan/smb_nfs_roundtrip.sh` (NFS slice)	Gated — reachabl bridge up export configure routes ov

Component	Feature	Category	Implementation	Wiring (reachal
	unmounted on every exit (§11.4.14)			

C. File transfer — AUTHORED, live round-trip operator-gated

Component	Feature	Category	Implementation	Wiring (reachable?)
FTP / FTPS	FTP passive directory-list + fetch over routed :21 + pinned PASV range (PASV must advertise the 10.x addr); FTPS explicit (`AUTH TLS`) / implicit (:990)	file-transfer / functional	`tests/vpn_lan/ftp_sftp_webdav.sh` (FTP slice)	Gated — reachable once bridge up + `HELIX_VPN_FTP_URL` set
SFTP	SFTP single-connection byte round-trip over routed :22 (key-based, `BatchMode` — fully autonomous, §11.4.98); sha256 match required	file-transfer / functional	`tests/vpn_lan/ftp_sftp_webdav.sh` (SFTP slice)	Gated — reachable once bridge up + `HELIX_VPN_SFTP_HOST` set
	WebDAV PROPFIND			Gated — reachable once bridge up +

Component	Feature	Category	Implementation	Wiring (reachable?)
WebDAV (via existing Squid)	through the EXISTING Squid (no new component) — expect `207 Multi-Status` + non-empty XML body; a real non-207 HTTP response ⇒ FAIL (fail-closed §11.4.68)	file-transfer / functional	`tests/vpn_lan/ftp_sftp_webdav.sh` (WebDAV slice)	`HELIX_VPN_WEBDAV_URL` set; traverses `HELIX_SQUID_PROXY` (default `:53128`)

D. Email — AUTHORED, live round-trip operator-gated

Component	Feature	Category	Implementation	Wiring (reachable
IMAP / IMAPS	IMAPS `LOGIN` + `LIST "" "*"` over implicit-TLS :993 (RFC 8314) — assert a real mailbox `* LIST` listing returns; `a1 NO/BAD` ⇒ FAIL	email / functional	`tests/vpn_lan/email_roundtrip.sh` (T4.1)	Gated — reachable bridge up `HELIX_MAIL_` `PASS` set; cr via in-proc `printf` stdi never argv logged (§11.4.10)
SMTP submission	Authenticated submission send over :465 implicit / :587 STARTTLS — a self-addressed round-trip probe token must be 2xx/250-accepted	email / functional	`tests/vpn_lan/email_roundtrip.sh` (T4.2)	Gated — reachable bridge up creds set

Component	Feature	Category	Implementation	Wiring (reachabl
POP3S	POP3S retrieve over implicit-TLS :995 — retrieve the just-sent T4.2 message and verify its unique token (a true send→retrieve round-trip); token not returned within retries ⇒ FAIL	email / functional	`tests/vpn_lan/email_roundtrip.sh` (T4.3)	Gated — reachable o bridge up + creds + a successful send
Open-relay guard (mandatory negative)	§4.3 MANDATORY negative test — an UNAUTHENTICATED relay to an EXTERNAL domain MUST be REFUSED: captured 4xx/5xx of the external RCPT ⇒ PASS; a captured 2xx acceptance ⇒ decisive FAIL (helix_proxy is NEVER an open relay / spam conduit); incomplete probe ⇒ honest SKIP (never fake-PASS the guard)	email / security	`tests/vpn_lan/email_roundtrip.sh` (T4.4)	Gated — reachable o bridge up + mail server present

E. Discovery reflector — AUTHORED design + test, remote deploy operator-gated

Component	Feature	Category	Implementation	Wiring (reachable?)
mDNS / DNS-SD / SSDP / WS-Discovery reflector	Remote-side multicast discovery reflector (Avahi `enable-reflector` for	discovery / functional	design `docs/design/vpn_lan_access/reflector_design.md`; test `tests/vpn_lan/discovery_reflect.sh`	Gated — enumeration reachable onc bridge up + re deployed (`HELIX_VPN_REF`

Component	Feature	Category	Implementation	Wiring (reachable?)
	mDNS 5353 / DNS-SD; SSDP relay for 1900; WS-Disc 3702) — multicast is subnet-local and NOT forwarded across the L3 VPN, so a remote-side reflector is required. Client test: `avahi-browse -rpt` surfaces a resolved (^=) remote service (SCORED); SSDP M-SEARCH count is supplementary/ non-scored			remote deploy operator-gated (§11.4.122)

F. Casting (Google Cast / DIAL) — AUTHORED, live round-trip operator-gated

Component	Feature	Category	Implementation	Wiring (reach
Chromecast eureka_info control	Routed `GET http://<ip>:8008/setup/eureka_info` over the L3 VPN — a real JSON `name` field is the device-identity evidence; HTTP 200 without a JSON `name`, or a non-200 answer ⇒ FAIL (fail-closed §11.4.68)	casting / functional	`tests/vpn_lan/chromecast_dial.sh` (T6.2)	Gated — reach bridge up + `HELIX_VPN_CAST_` control routes TCP

Component	Feature	Category	Implementation	Wiring (reacl
CASTV2 liveness (§11.4.107)	Cast status *transition* observed across two reads (advancing state, NOT a single frozen frame) via an operator-supplied status command; two DISTINCT reads ⇒ transition PASS; identical reads (device idle) ⇒ honest `operator_attended` SKIP (a transition needs media playing), never a fake PASS	casting / functional (liveness)	`tests/vpn_lan/chromecast_dial.sh` (T6.3)	Gated — reach bridge up + `HELIX_VPN_CAST_` set
DIAL discovery + control	DIAL discovery via the Phase-5 reflector (SSDP `M-SEARCH`) + `_googlecast._tcp` mDNS; HTTP control routes over the VPN as ordinary unicast TCP (the routable "cast to a remote display" alternative to Miracast)	casting / functional	`tests/vpn_lan/chromecast_dial.sh` (T6.1 discovery) + `reflector_design.md`	Gated — disco depends on the (E); control rou bridge up

G. Device bridge (ADB over VPN) — AUTHORED; flash operator-blocked

Component	Feature	Category	Implementation	Wiring (reachable?)
ADB connect / shell / getprop	`adb connect <host>:5555` over the routed VPN (no proxy hop — recon 4) + central adb-server model; assert OUR serial reaches `device` state then `adb -s <serial> shell getprop ro.product.model` returns real content; reachable-but-unusable state (offline/unauthorized) ⇒ FAIL. Safety: touches ONLY the env-configured serial — never `kill-server`, never a blanket disconnect, never any other/operator/lava-* device (§11.4.174); `adb disconnect <our-serial>` in the cleanup trap	device-bridge / functional	`tests/vpn_lan/adb_over_vpn.sh` (T7.1/T7.3)	Gated — reachable once bridge up + `HELIX_VPN_ADB_H` set
ADB flash (usbip USB-over-IP)	Honest boundary (§11.4.6): `fastboot` is USB-level and network fastboot is USB-bound (recon 4 FACT); the routable path is `usbip` from a	device-bridge / operator-safety	`tests/vpn_lan/adb_over_vpn.sh` (T7.4)	Gated — real-device flash requires explici operator authorization

Component	Feature	Category	Implementation	Wiring (reachable?)
	remote host with the device attached. The test NEVER flashes and NEVER invokes fastboot/usbip — it records the boundary and SKIPs `operator_attended`; any real-device flash is operator-gated (§11.4.133 + §11.4.122)			

H. Miracast — Won't-fix, structurally-impossible

Component	Feature	Category	Implementation	Wiring (reachable?)	Re
Miracast over L3 VPN	Won't-fix — structurally-impossible (§11.4.112): Miracast rides Wi-Fi Direct / 802.11 P2P at Layer 2 with no routable IP hop for an L3 VPN to carry; cited Wi-Fi-Alliance + Wi-Fi-Direct evidence. The routable "cast to a remote display" capability is	honest-boundary / structural verdict	`docs/design/vpn_lan_access/miracast_verdict.md` (cited spec text IS the artifact)	N/A — out of scope by physics; positive capability lives in §F (Cast/DIAL)	Th doc del rec onl NE evi the pla cor cha (§1 §1

Component	Feature	Category	Implementation	Wiring (reachable?)	Re
	delivered by Google Cast / DIAL (§F). NO fake traversal test exists or will be authored (a green Miracast-over-VPN result would be fabricated)				

I. Documentation & verification infrastructure

Component	Feature	Category	Implementation	W
Operator bridge-setup guide	End-user manual — overview, one-line routing map, the 6-var contract, setup steps, per-protocol matrix, security notes, FAQ; HTML+PDF exports §11.4.65, §11.4.168 visually validated (no raw diagram source leaking)	docs / operator	`docs/guides/vpn_lan_bridge_setup.md`	Y d d e e f
VPN-LAN Challenge runner	Runnable anti-bluff harness — runs `svord_doctor.sh` preflight then each `tests/vpn_lan/*.sh`, maps exit codes (doctor 0=UP/ 2=SKIP/ 3=MISCONFIGURED; tests 0=no-FAIL/ 1=FAIL), tallies PASS/	verification / Challenge (§11.4.27)	`challenges/scripts/run_vpn_lan_challenges.sh`	Y c r i c c p - h t

Component	Feature	Category	Implementation
	FAIL/SKIP with per-item logs under host-safety caps (`GOMAXPROCS=2 nice ionice`, §12.6). Bridge-down ⇒ every item honest SKIP, exit 0 (never a fake PASS); a not-yet-present test ⇒ SKIP, never FAIL		
HelixQA `vpn_lan.yaml` bank	HelixQA test bank — one case per protocol (`VLA-BRIDGE-PREFLIGHT` / `VLA-SMB` / `VLA-NFS` / `VLA-FTP` / `VLA-SFTP` / `VLA-WEBDAV` / `VLA-EMAIL` / `VLA-CAST` / `VLA-ADB`) dispatching `ActionTypeShell` to the real protocol test so the case verdict IS the real test verdict; documents the exit-code→verdict mapping	verification / HelixQA (§11.4.27)	`tools/helixqa/banks/vpn_lan.yaml`

§11.4.169 test-type coverage note (honest boundary)

The autonomously-provable layers today are: **unit / config** (bridge-contract resolution + set-u safety), the **standing-suite gate** (`test_vpn_lan_bridge` — 3-verdict discrimination + §11.4.115 UP-stub teeth, GREEN), the **Challenge runner** (runnable NOW, honest bridge-down all-SKIP tally), and the **structural verdict** (Miracast, cited). The **integration / e2e / full-automation / security / stress+chaos** live layers per protocol are AUTHORED with evidence-gating + wrong-answer-FAIL teeth but **honestly SKIP** until the operator connects the svord bridge (§11.4.3) — they are never counted as PASS on a down bridge. HelixQA autonomous-session coverage is **run-blocked** (6 un-vendored siblings, §11.4.3), surrogated by the Challenge runner. No metadata-only / config-only / absence-of-error PASS is recorded anywhere in this ledger (§11.4 / §11.4.1 / §11.4.69).

Honest boundary (§11.4.6)

What is PROVEN NOW, autonomously, with committed evidence: the decoupled env-var bridge contract, the svord-doctor 3-verdict preflight with §11.4.115 teeth in the GREEN standing suite, the honest-SKIP anti-bluff gate across every protocol test (bridge-down ⇒ SKIP + exit 0 + zero `^PASS:` lines — never a fake PASS), the wrong-answer-FAIL teeth (WebDAV non-207, SFTP/NFS/SMB sha-mismatch, open-relay external-RCPT-accepted, ADB not-`device`-state, Cast non-200/no-name), the runnable Challenge harness, the operator guide, and the Miracast structural-impossibility verdict. What is **operator-gated** and therefore SKIP/OPERATOR-BLOCKED until unblocked: every **live** protocol round-trip (real bytes, sha256, mailbox content, eureka JSON, device serials — needs the svord bridge up), the multicast-reflector remote deployment (§11.4.122), any real-device flash (§11.4.133), and the autonomous HelixQA session (6 un-vendored siblings). No live capability is marked PASS without captured evidence on a genuinely-up bridge, and no feature is "done" for its live capability until its runtime signature verifies on a live bridge (§11.4.108) and it crosses the §11.4.169 test-type matrix + independent review (§11.4.142). This ledger does not substitute for the §11.4.40 full-suite retest before any release tag.

Feature tally: 4 rows PASS-now (bridge contract · svord-doctor preflight · operator guide · Challenge runner) · 16 rows operator-gated SKIP/OPERATOR-BLOCKED (all live protocol round-trips + reflector deploy + ADB flash + HelixQA run) · 1 row Won't-fix (Miracast, structurally-impossible §11.4.112).